

Fold-by-Fold Performance: Why AR+News Helps in Some Periods and Not Others

7 test folds across Feb 2022 – Feb 2024 | Primary metric: PR-AUC | All values from actual results

Overview: All 7 Folds at a Glance

Fold	Period	n	Prevalence	AR PR-AUC	AR+News PR-AUC	Delta
Fold 1	Feb 2022	263	20.2%	0.889	0.899	+0.009
Fold 2	Jun 2022	297	22.2%	0.907	0.904	-0.003
Fold 3	Oct 2022	263	27.0%	0.909	0.886	-0.024
Fold 4	Feb 2023	341	29.9%	0.862	0.885	+0.023
Fold 5	Jun 2023	341	33.7%	0.714	0.887	+0.174
Fold 6	Oct 2023	343	42.3%	0.851	0.849	-0.003
Fold 7	Feb 2024	300	28.0%	0.710	0.788	+0.079

News helps in 4 of 7 folds (delta > 0). The largest gain is Fold 5 (Jun 2023, +0.174) where AR-Only collapses to near-random performance. The three negative folds share a common pattern: low onset count, strong chronic dominance, and stable IPC history that AR alone handles well.

Key Metrics

Test observations (n):	263
Crisis cases (positives):	53 (20.2% prevalence)
AR-Only PR-AUC:	0.8894
AR+News PR-AUC:	0.8987
Delta PR-AUC:	+0.0093
AR-Only ROC-AUC:	0.9601
AR+News ROC-AUC:	0.9544

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	11	+0.524	9%	73%
Chronic	42	+0.107	100%	100%
Recovery	5	+0.168	100%	100%
Stable	205	+0.215	0%	25%

What Drove This Fold's Result

"News dramatically lifts onset recall; AR history is thin for new deteriorations"

Feb 2022 is the earliest test fold — IPC history is sparse for newly deteriorating districts. The 11 onset cases are poorly covered by lagged IPC signal: AR-Only achieves just 9% onset recall at threshold 0.5. AR+News lifts this to 73% (+0.524 mean delta probability on onset cases) by picking up conflict and displacement coverage that preceded the assessments. The 42 chronic cases also benefit slightly (delta +0.107). Stable districts are lifted as well (+0.215), suggesting news volume adds general upward pressure on predictions, but the net PR-AUC gain of +0.009 is positive and ROC-AUC moves from 0.960 to 0.954 — a modest trade-off where precision improvements on onset are partially offset by false positives among stable districts.

Key Metrics

Test observations (n):	297
Crisis cases (positives):	66 (22.2% prevalence)
AR-Only PR-AUC:	0.9070
AR+News PR-AUC:	0.9040
Delta PR-AUC:	-0.0030
AR-Only ROC-AUC:	0.9775
AR+News ROC-AUC:	0.9702

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	2	-0.566	100%	0%
Chronic	64	-0.053	100%	97%
Recovery	16	-0.068	100%	100%
Stable	215	-0.112	25%	2%

What Drove This Fold's Result

"AR-Only peaks; news adds noise on a highly stable, predictable period"

Jun 2022 is the best fold for AR-Only (PR-AUC 0.907, ROC-AUC 0.978). With only 2 onset cases, there is almost no new-crisis signal for news to detect — and news actually hurts these 2 cases sharply (delta −0.566, dropping recall from 100% to 0%). The 64 chronic cases are near-perfectly captured by lagged IPC history alone, and news lowers their probabilities by −0.053 on average. All regimes see a negative news delta. Delta PR-AUC = −0.003. When history is a near-perfect guide and no new crises are emerging, adding news introduces more noise than signal.

Key Metrics

Test observations (n):	263
Crisis cases (positives):	71 (27.0% prevalence)
AR-Only PR-AUC:	0.9091
AR+News PR-AUC:	0.8855
Delta PR-AUC:	-0.0236
AR-Only ROC-AUC:	0.9642
AR+News ROC-AUC:	0.9473

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	5	-0.250	20%	0%
Chronic	66	-0.095	100%	92%
Recovery	13	-0.080	100%	100%
Stable	179	-0.067	4%	0%

What Drove This Fold's Result

"News hurts both onset and chronic; a period where coverage contradicts IPC trajectory"

Oct 2022: 5 onset cases with 20% recall for AR-Only — already weak. News reverses this to 0% (delta -0.250), actively moving onset cases away from threshold. Chronic dominates (66 cases) and news pulls their probabilities down by -0.095 on average, also costing recall (100% to 92%). All four regimes show negative news deltas. Delta PR-AUC = -0.024, the largest negative fold. This period illustrates the failure mode: when GDELT coverage patterns do not align with impending IPC deteriorations — either because crises are geographically diffuse, temporally displaced, or in areas with limited English-language coverage — news features actively mislead the model.

Key Metrics

Test observations (n):	341
Crisis cases (positives):	102 (29.9% prevalence)
AR-Only PR-AUC:	0.8622
AR+News PR-AUC:	0.8852
Delta PR-AUC:	+0.0230
AR-Only ROC-AUC:	0.9289
AR+News ROC-AUC:	0.9489

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	26	-0.020	8%	0%
Chronic	76	+0.065	84%	92%
Recovery	11	+0.105	100%	100%
Stable	228	-0.015	1%	0%

What Drove This Fold's Result

"News rescues a period of rising prevalence by lifting chronic detection"

Feb 2023 sees a jump to 102 crisis cases (30% prevalence) and AR-Only PR-AUC drops to 0.862. AR+News recovers to 0.885 (+0.023). The gain is driven primarily by the 76 chronic cases: news adds +0.065 mean delta probability, lifting chronic recall from 84% to 92%. The 26 onset cases are not helped (delta -0.020, recall stays near 0%) — news does not detect these new deteriorations. ROC-AUC improves substantially (0.929 to 0.949), indicating that the probability ranking across all districts improves even where threshold-level recall does not change.

Key Metrics

Test observations (n):	341
Crisis cases (positives):	115 (33.7% prevalence)
AR-Only PR-AUC:	0.7136
AR+News PR-AUC:	0.8871
Delta PR-AUC:	+0.1735
AR-Only ROC-AUC:	0.7578
AR+News ROC-AUC:	0.9515

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	43	+0.228	2%	9%
Chronic	72	-0.017	100%	97%
Recovery	19	-0.110	100%	84%
Stable	207	-0.085	9%	0%

What Drove This Fold's Result

"Largest gain: AR-Only collapses, news rescues the model on onset cases"

Jun 2023 is the standout fold. AR-Only PR-AUC falls to 0.714 and ROC-AUC to 0.758 — a near-random classifier for this period. 43 onset and 72 chronic cases at 33.7% prevalence overwhelm the lagged IPC signal. AR+News recovers to PR-AUC 0.887 and ROC-AUC 0.952 — a +0.174 PR-AUC gain. News lifts onset recall from 2% to 9% (delta +0.228) — a modest absolute recall lift but a massive improvement in probability ranking for onset districts. Chronic cases are nearly unaffected (delta −0.017, recall 100% to 97%). This is the strongest evidence that GDELT coverage carries genuine pre-crisis signal: it partially compensates for a period where IPC history completely fails.

Key Metrics

Test observations (n):	343
Crisis cases (positives):	145 (42.3% prevalence)
AR-Only PR-AUC:	0.8513
AR+News PR-AUC:	0.8487
Delta PR-AUC:	-0.0026
AR-Only ROC-AUC:	0.8840
AR+News ROC-AUC:	0.8966

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	53	+0.019	23%	23%
Chronic	92	-0.029	100%	99%
Recovery	18	+0.000	100%	100%
Stable	180	-0.012	1%	1%

What Drove This Fold's Result

"Effectively neutral; onset recall unchanged, chronic barely moves"

Oct 2023 has the highest prevalence in the test set (42.3%, 145 positives). Both models perform moderately: AR-Only PR-AUC 0.851, AR+News 0.849 (delta −0.003). Despite 53 onset cases, news adds almost no delta to onset (+0.019, recall flat at 23%). The 92 chronic cases are already at 100% recall under AR-Only and news barely changes this (−0.029 delta, 99% recall). At very high crisis prevalence, the AR lag features are already strong and news cannot improve on them. The model is operating close to its ceiling for chronic detection and news neither helps nor hurts meaningfully.

Key Metrics

Test observations (n):	300
Crisis cases (positives):	84 (28.0% prevalence)
AR-Only PR-AUC:	0.7097
AR+News PR-AUC:	0.7884
Delta PR-AUC:	+0.0787
AR-Only ROC-AUC:	0.8759
AR+News ROC-AUC:	0.8668

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	36	-0.034	3%	3%
Chronic	48	+0.038	94%	92%
Recovery	16	-0.383	100%	38%
Stable	200	-0.013	2%	2%

What Drove This Fold's Result

"Meaningful ranking improvement; PR-AUC gains +0.079 despite 0% onset recall at threshold"

Feb 2024: 36 onset cases, 3% recall for both models at threshold 0.5 — neither model is confident enough to flag most new crisis districts. Yet PR-AUC improves from 0.710 to 0.788 (+0.079), indicating that news features substantially improve the probability ranking even without pushing cases above the classification threshold. Chronic cases show a small positive news effect (+0.038 delta). The large negative delta on recovery cases (−0.383, recall dropping from 100% to 38%) is a localised cost. The improvement in PR-AUC ranking is real and meaningful — it would translate to actionable early warning under a lower operating threshold.